

FACULTY OF ECONOMIC AND MANAGEMENT SCIENCES/ FAKULTEIT EKONOMIESE EN
BESTUURSWETENSKAPPE

DEPARTMENT OF ECONOMICS/ DEPARTEMENT EKONOMIE

Module Code: 12084-114

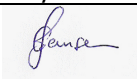
ECONOMICS 114/ EKONOMIE 114

ASSESSMENT A3 / ASSESSERING A3

CLOSED BOOK/ TOE BOEK

PAPER 1 OF 1 / VRAESTEL 1 VAN 1

DATE/ DATUM	14 JUNE 2024/14 JUNIE 2024
DURATION/TYDSDUUR	3 HOURS / 3URE
TOTAL OF MARKS/ TOTALE PUNTE	100
EXAMINERS/EKSAMINTORE:	DR C LINKS PROF S DU PLESSIS MS O EZEObI MR M MEYER DR K MALINDI MR J PRETORIUS DR D SHEPHERD DR P VORSTER
INTERNAL MODERATOR/ INTERNE MODERATOR	DR E MOSES

Chairperson / Voorsitter:	Co-ordinator / Sameroeper:
Prof AI Jansen 	Dr C Links

INSTRUCTIONS

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.

ANSWER **SECTION A's** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY **DARKLY COLOURING IN THE CIRCLE** NEXT TO THE NUMBER OF THE STATEMENT.

ALSO ANSWER **SECTION B's** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY **DARKLY COLOURING IN THE CIRCLE** NEXT TO THE NUMBER OF THE STATEMENT.

ANSWER **SECTION C's** QUESTIONS IN THE ANSWER BOOK PROVIDED. START EVERY QUESTION ON A NEW PAGE. SHOW ALL CALCULATIONS.

KEY:

AP = Average product
AC = Average cost
BL = Budget line
D = Demand
DWL = Deadweight loss
FF = Feasible frontier
GDP = Gross domestic product
IC = Indifference curve
K = Capital
L = Labour
MB = Marginal benefit
MC = Marginal cost
MEB = Marginal external benefit
MP = Marginal product
MPB = Marginal private benefit
MPC = Marginal private cost

MR = Marginal revenue
MRS = Marginal rate of substitution
MRT = Marginal rate of transformation
MSB = Marginal social benefit
MSC = Marginal social cost
P = Price
PF = Production function
Q = Quantity
 Q_d = Quantity demanded
 Q_s = Quantity supplied
S = Supply
TC = Total cost
TP = Total product
TR = Total revenue
w = Wage
Y = Income

SECTION A
TRUE OR FALSE STATEMENTS

[10 marks: 1 mark for a correct answer and - ½ for an incorrect answer]

1. The GDP per capita of China was \$7 636 in 2014 and \$8 016 in 2015. Based on these figures, the growth rate of GDP per capita of China between 2014 and 2015 (measured to two decimal places) was 4.98%.
 - a) True
 - b) False

2. According to Malthus, the population size will increase if living standards increase, but then the average product of labour will eventually decline.
 - a) True
 - b) False

3. After Melody's wage increased, she chose to work less because her income effect on free time dominated her substitution effect on free time.
 - a) True
 - b) False

4. A Nash equilibrium is always a dominant strategy equilibrium.
 - a) True
 - b) False

5. The Nash equilibrium of a prisoner's dilemma game represents a social dilemma.
 - a) True
 - b) False

6. Previously, a corrupt government official frequently accepted bribes, driven solely by personal financial interests. Due to the establishment of stronger institutions that effectively mitigate corrupt practices, the official can no longer solicit or accept bribes. This change represents a Pareto improvement.
 - a) True
 - b) False

7. In the labour discipline model, it is assumed that the firm must pay the worker exactly his/her reservation wage to create an incentive for him/her to work hard.
 - a) True
 - b) False

8. In a perfectly competitive market, the process of entry and exit of firms will end when marginal cost is equal to average cost.
 - a) True
 - b) False

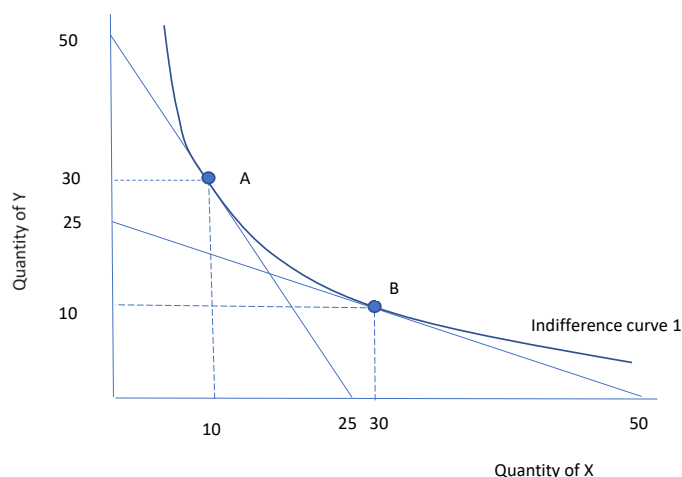
9. The MR curve for a monopolist is also its demand curve.
 - a) True
 - b) False

10. When VW car manufacturer is forced to consider the social cost of production, it must add the costs that are incurred by others affected by their production, to its private costs of production.
 - a) True
 - b) False

SECTION B
MULTIPLE CHOICE QUESTIONS

[40 marks: 2 marks for a correct answer and – ¼ for an incorrect answer]

11. Which one of the following statements is **INCORRECT**?
- An economic system is a way of organising the production and distribution of goods and services in an entire economy.
 - An authoritarian government can never exist in a capitalist economic system.
 - There remains a role for governments in modern capitalist systems.
 - Markets existed before capitalism.
12. A change in the relative price of two production inputs will affect the
- marginal product of labour.
 - slope of the isocost line.
 - slope of the production function.
 - marginal rate of substitution.
13. The diagram below shows the budget constraint and an indifference curve of a consumer of two goods, X and Y.

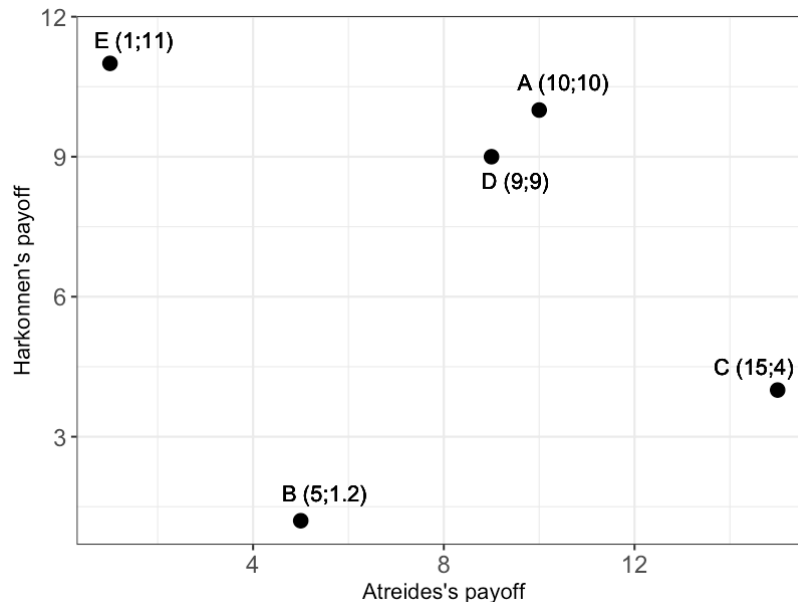


At point A, the MRS is _____ and at point B the MRS is _____.

- 0.33; 3
- 0.5; 2
- 2; 0.5
- impossible to calculate with the information given.

14. Consider a consumer with a fixed budget that chooses to consume a combination of two normal goods, chicken and chips. The consumer's budget line is plotted with chicken on the horizontal axis. When the price of chicken increases:
- a) the substitution effect on the consumption of chicken is positive, as the consumer will have to increase his/her expenditure on chicken due to its higher price, *ceteris paribus*.
 - b) the income effect on the consumption of chips is negative, *ceteris paribus*.
 - c) the consumer will consume less chips because the substitution effect dominates the income effect, *ceteris paribus*.
 - d) the consumer will consume more chicken because the income effect dominates the substitution effect, *ceteris paribus*.
15. Melinda is a graphic designer who charges R400 per hour to produce social media marketing adverts for a leading company. She spends 8 hours per day sleeping and allocates the remaining 16 hours between work and free time. If we plot Melinda's budget constraint with free time on the horizontal axis and consumption on the vertical axis, _____.
- a) her MRS at her best affordable point is greater than R400.
 - b) her MRS at her best affordable point is smaller than R400.
 - c) the relative price of consumption is R400.
 - d) her per hour opportunity cost of free time is R400.
16. Which one of the following statements is **CORRECT**?
- a) A dominant strategy equilibrium always arises when both players play their best responses.
 - b) A dominant strategy equilibrium implies that each player changes their strategy based on the other player's strategy.
 - c) The Nash equilibrium is a dominant strategy equilibrium when players choose a strategy independently of the other players.
 - d) A strategy is dominant when a player's payoffs are greater in size than the other players' payoffs.

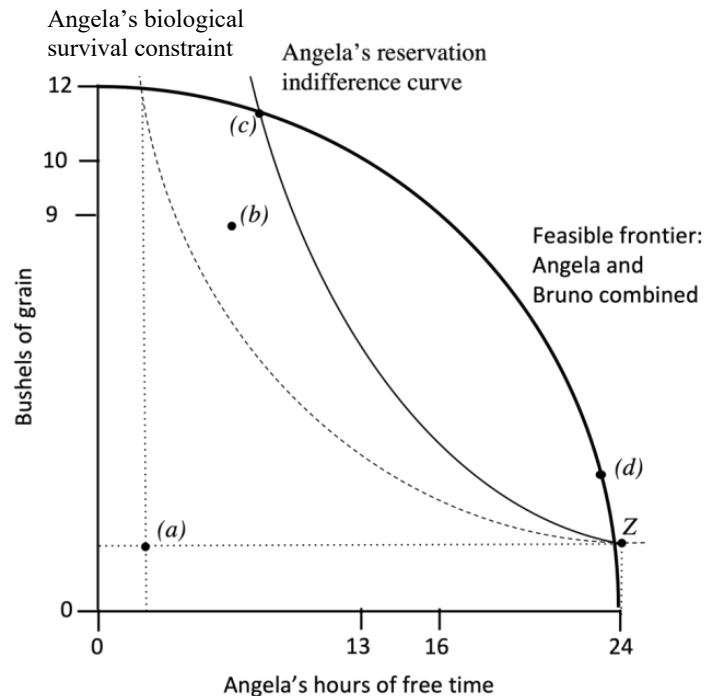
17. Consider the following figure showing the payoff allocations for two persons, Atreides and Harkonnen:



Which of the following statements is/are **CORRECT**?

- I. Allocation B is only Pareto dominated by allocations A and D.
 - II. Allocation A is Pareto efficient.
 - III. Allocation E and C are Pareto efficient.
 - IV. Moving from allocation C to D results in a Pareto improvement.
- a) I, II, and III
 - b) Only II
 - c) II and III
 - d) II, III, and IV
18. Zephyr works for a window washing company and earns a wage of R75 per hour. He works a 40-hour work week. He finds his job quite boring, and has a disutility of effort of R25 per hour. The government provides an unemployment benefit equivalent to R30 per hour. Which one of the following statements is **INCORRECT**?
- a) To calculate the cost of job loss, Zephyr needs to compare the value of his current job to the value of his reservation option.
 - b) If Zephyr is offered another window washing job that pays R30 per hour, he will definitely take this job.
 - c) If Zephyr's employer provided health insurance and a travel allowance, Zephyr's employment rent would be higher.
 - d) Zephyr's disutility of effort reduces his employment rent.

19. Bruno is a landowner and Angela is a farmer who pays a share of her grain output to Bruno for the use of his land. Which of the allocation(s) indicated represent a Pareto improvement over Z ?



- a) (c) and (d)
 b) only (c)
 c) only (d)
 d) None of the allocations
20. Economies of scale necessarily exist when:
- a) the firm's decision to hire additional labour does not result in an increase in wages.
 b) a firm is large and diversified.
 c) the average cost of producing a unit of output falls as output increases.
 d) a firm is small and very specialised.
21. A monopoly is best defined as a firm operating in an industry where barriers prevent
- a) other firms from selling that good or service.
 b) product uniqueness.
 c) many buyers from purchasing the good or service.
 d) price-setting.

22. A monopolist's demand and marginal cost functions are given by:

$$Q = 400 - 5P$$

$$MC = 20 + 0,2Q$$

What is the monopolist's profit-maximising output?

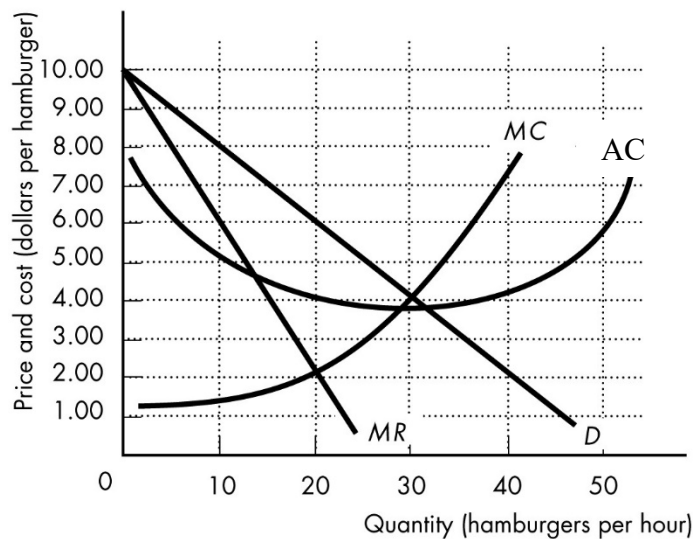
- a) 250
 - b) 150
 - c) 500
 - d) 100
23. A price-setting firm can sell 100 units of its product at a price of R250. However, to sell 101 units they would need to lower the price to R248. What is the marginal revenue of the 101st unit?
- a) -R2
 - b) R2
 - c) R48
 - d) -R48
24. A binding price floor and a binding price ceiling both have the result that
- a) a smaller quantity, relative to the equilibrium quantity, will be sold.
 - b) a larger quantity, relative to the equilibrium quantity, will be sold.
 - c) producer surplus increases.
 - d) consumer surplus increases
25. "Charlie's Chimps" is a perfectly competitive firm that produces soft toy chimpanzees for children. The market price of a chimpanzee is R10 and Charlie produces 100 chimpanzees at a marginal cost of R9. The firm _____
- a) is maximising its profit.
 - b) will maximise its profit if it produces more than 100 chimpanzees.
 - c) will maximise its profit if it lowers the price to R9 a chimpanzee.
 - d) will maximise its profit if it produces fewer than 100 chimpanzees.

26. The demand function for a price-setting firm is $Q = 100 - \frac{1}{80}P$. When $Q = 13$, MR is equal to _____.
- R5 920
 - R6 240
 - R1 100
 - R6 400
27. Suppose that the average cost of a particular firm is given by:

$$AC = 2Q - 15 + 20\,000/Q$$

The average cost is at a minimum when quantity equals _____.

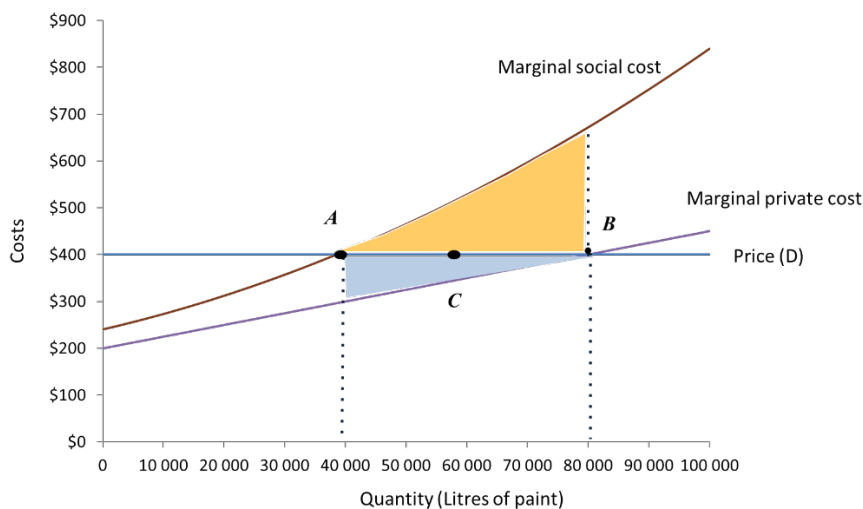
- 150
 - 100
 - 200
 - 85
28. The figure below represents *Ron's Hamburger Joint*, the only restaurant in town. Ron's AC, MC, MR and demand curves are indicated. All prices and costs are given in \$.



If *Ron's Hamburger Joint* produces the profit-maximising quantity of hamburgers, their economic profit will be _____ per hour.

- \$40
- \$30
- \$20
- \$120

29. Suppose that private solutions have failed to encourage brick manufacturing factories to lower their emissions, resulting in market failure and deadweight loss. Governments can regulate the market to produce a Pareto efficient output by
- banning brick manufacturing entirely.
 - investing in the expansion of brick factories so that emissions are more evenly distributed.
 - regulating the quantity of emissions per factory.
 - taxing the production of bricks.
- only I
 - I, II and IV
 - I, III and IV
 - only III and IV
30. The market for paint is competitive and unregulated. Suppose a paint factory, Ray's Paint, disposes of its dirty water into a nearby river. This pollution is slowly killing off the fish stock, which is harming the local fishery, Raymond's Big Catch.



Which one of the following statements is **CORRECT**?

- Ray's Paint chooses to produce 40 000 litres of paint, because this is where the $MPB = MSC$.
- Ray's Paint will produce 80 000 litres of paint, because this is where $MPC = P$.
- If Ray's Paint owns the river, there is no way to deal with the cost caused by the pollution.
- Raymond wants to do what is right for the environment and closes his paint business.

SECTION C
DESCRIPTIVE QUESTIONS
[50 marks]

QUESTION 31 [2 marks]

A 2019 International Monetary Fund (IMF) research paper found that policies targeted at redistributing unpaid household work across men and women could increase GDP by 4 percent. Discuss briefly why this positive effect could occur. [2]

QUESTION 32 [5 marks]

A winemaker is faced with five technologies (A, B, C, D, E) that use different combinations of workers and carts of grapes to produce a given level of output, as indicated in the table below. Workers are paid a wage of R900 and a cart of grapes costs R3 000.

Technology	Number of workers	Carts of grapes
A	10	4
B	4	5
C	5	7
D	8	3
E	1	8

- 32.1 Which technologies are dominated? [2]
- 32.2 Give the equation of the isocost line that represents a total cost of production of R300 000 if carts of grapes are plotted on the vertical axis. [2]
- 32.3 What does the slope of the isocost line in 32.2 represent? [1]

QUESTION 33 [2 marks]

Use the information provided in the table below to answer the questions that follow.

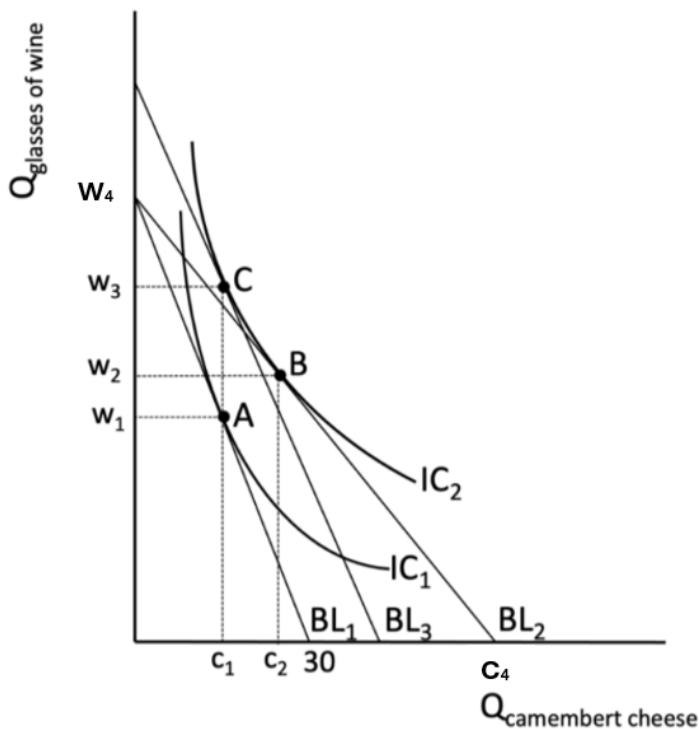
Labour employed (L)	0	15	30	45	60	75
Output (Q)	0	300	600	750	800	900

33.1 What is the MP of the 30th unit of labour? [1]

33.2 What is the AP of the 30th unit of labour? [1]

QUESTION 34 [5 marks]

Simon has an income of R3 000 that he spends on two goods: wine and camembert cheese. Wine costs R30 per glass, and a camembert cheese costs R100. Several of Ben's budget lines and indifference curves are indicated in the figure below (not drawn to scale).



The price of a camembert cheese decreases from R100 to R60. Use the information provided, and determine the value of:

34.1 W_4 [1]

34.2 C_4 [1]

34.3 MRS at C [3]

QUESTION 35 [3 marks]

The following payoff matrix applies to a game played by two players, Player 1 and Player 2, who can choose to play strategy A or strategy B:

		Player 2	
		A	B
Player 1	A	2, 2	-4, 4
	B	4, -4	-8, -8

35.1 What is the Nash equilibrium of this game? [2]

35.2 Does Player 1 and/or Player 2 have a dominant strategy? Explain your answer. [1]

QUESTION 36 [5 marks]

Bronwyn, Nonso, and Leonard are discussing how to share three chocolate ice-creams and three strawberry ice-creams. Leonard likes chocolate and strawberry ice-creams equally, but Nonso prefers chocolate to strawberry ice-cream, and Bronwyn prefers strawberry to chocolate ice-cream.

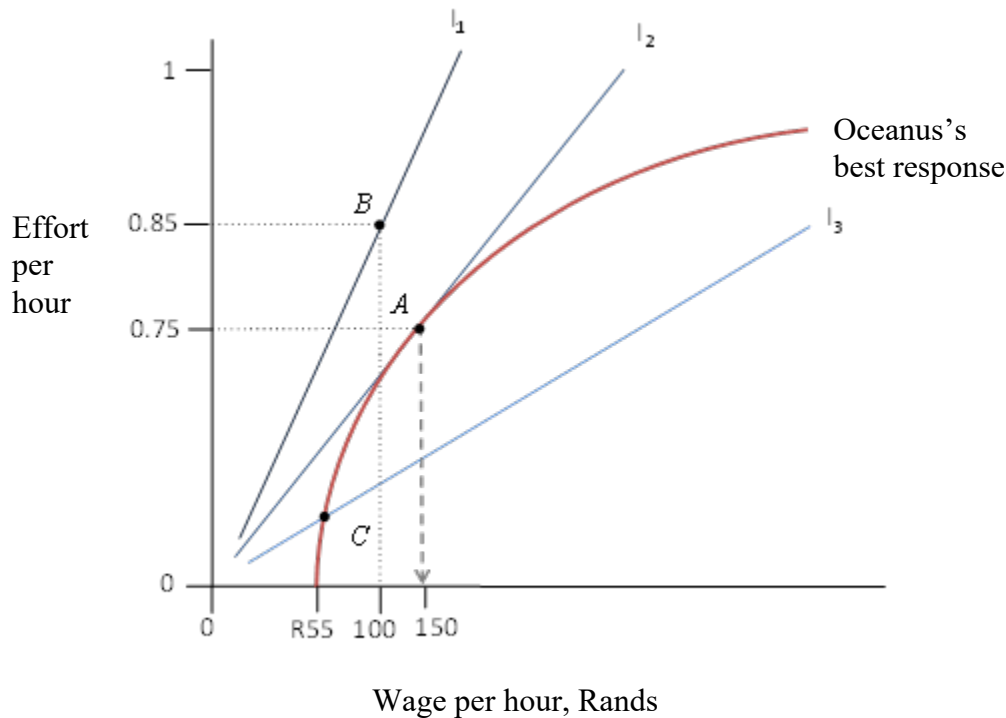
36.1 Is an allocation in which Leonard gets two strawberry ice-creams, Nonso gets a chocolate ice-cream and a strawberry ice-cream, and Bronwyn gets two chocolate ice-creams Pareto efficient? Explain. [3]

Johan, who also likes chocolate and strawberry ice-creams equally, steals all the ice creams for himself.

36.2 Is this allocation in which Johan gets all the ice-creams Pareto efficient? Explain. [2]

QUESTION 37 [4 marks]

The firm "Titan Forge" is making a decision on what wage to pay their employee, Oceanus.



- 37.1 Which one of the isocost curves I_1 , I_2 or I_3 represents Titan Forge's highest cost of effort? Why? [1]
- 37.2 What wage would Titan Forge pay in order to maximise its profit? [1]
- 37.3 Which of points A, B and C would be feasible for Titan Forge? Briefly explain your answer. [2]

QUESTION 38 [7 marks]

The equations below represent the market demand and market supply for eggs:

$$Q_d = 10\,000 - 1000P$$

$$Q_s = 4\,000 + 1000P$$

Suppose the government imposes a price floor of R5 per egg.

- 38.1 Is there a shortage or a surplus in this market following the introduction of the price floor, and what is the size thereof? [3]

- 38.2 How does the introduction of a price floor affect the size of consumer surplus? No calculations are required. [1]
- 38.3 Draw a fully labelled graph that illustrates the deadweight loss in the market for eggs as a result of the price floor. Your graph does not need to be drawn to scale. [3]

QUESTION 39 [3 marks]

Every year during the annual budget speech, the Minister of Finance increases excise taxes on cigarettes. Explain, using your knowledge of price elasticity of demand, the rationale behind this decision. [3]

QUESTION 40 [5 marks]

The Golden Tuckshop asks R12,50 per can of Pepsi. At this price the quantity demanded is 500. If Golden Tuckshop lowers the price to R10 per can of Pepsi, the quantity demanded rises to 750.

- 40.1 Calculate Golden Tuckshop's price elasticity of demand for Pepsi when the price charged is R12,50. [3]
- 40.2 Given your answer in 40.1, would you advise that the shop owner should consider lowering the price of a can of Pepsi? Motivate your answer. [2]

QUESTION 41 [5 marks]

Your roommate complains about the cost of living and says he wishes the government would create a maximum price for milk of R12 per litre. Assume the market-clearing price is R13 per litre of milk.

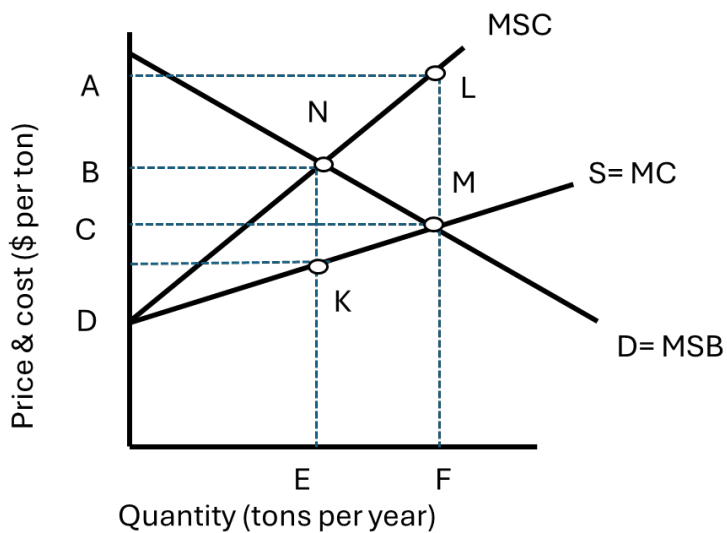
- 41.1 Explain to your roommate the welfare effects of this price ceiling on milk. No graph is required. [4]

Suppose that farm conditions improve in the following year, which results in a market equilibrium price of R11 per litre of milk.

- 41.2 How would you classify the price ceiling of R12 per litre of milk given this change in the market? [1]

QUESTION 42 [4 marks]

The figure below shows the market for coal. The production of coal leads to pollution that imposes an external cost.



42.1 What letter in the above figure represents the following:

42.1.1 Equilibrium price in an unregulated, competitive market. [1]

42.1.2 Equilibrium quantity in an unregulated, competitive market. [1]

42.2 What area represents the deadweight loss of the unregulated, competitive market outcome? [1]

42.2 What distance represents the size of the Pigouvian tax per ton that will be charged to produce the Pareto efficient outcome? [1]